DEEPESH PATIL

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Summary

AI/Data Engineer with hands-on experience building GenAI applications and data pipelines using Databricks and PySpark. Experienced in LLM evaluation, MLflow-based observability, and synthetic healthcare data generation. Also built and deployed full-stack Python applications with a focus on data validation, security, and reliable delivery.

Technical Skills

Languages: Python, SQL, Bash, Go, C/C++
AI/Data: PySpark, Databricks, Delta Lake, Unity Catalog, MLflow, Langfuse, LLM evaluation, RAG, FAISS
Backend/Web: FastAPI, Flask, React, TypeScript, REST APIs
Cloud/Delivery: Docker & Kubernetes, AWS, Git, GitHub Actions, Databricks Asset Bundles, Linux

Experience

Abacus Insights *Jun 2026 – Oct 2026*
AI Engineer

- Built and deployed a **Databricks** App for Genie NL2SQL evaluation, capturing **MLflow** traces in **Unity Catalog** and scoring outputs with schema-grounding checks and MLflow judges.
- Developed a deterministic **PySpark** synthetic eligibility-data generator (8 scenario types, 25 quality gates, idempotent **Delta** writes), producing 752K eligibility spans for 422K synthetic members.
- Built a synthetic Payment Integrity case pipeline with deterministic IDs and nine validation gates; delivered 17 approved cases.
- Set up **Databricks Asset Bundle** deployment for multi-workspace delivery with CI packaging checks.
- Compared **MLflow** and **Langfuse** observability on 1,145 benchmark traces and documented recommendations.

Supervised Project *Apr 2026 – May 2026*

- Built a claim-denial prevention prototype on synthetic data using **XGBoost**, **SHAP**, **FastAPI**, **Next.js**, and a **Databricks** medallion pipeline.

Training Program *Feb 2026 – Mar 2026*

- Completed AI/Data Engineering training covering **Python**, **SQL**, **AWS** fundamentals, and **Databricks**.

Saints Peter & Paul Roman Catholic Church *Jul 2025 – Aug 2025*
Full-Stack Developer (Volunteer)

- Built and deployed a self-hosted **CRM & DBMS** using **React** and **MySQL** in the church's environment, enabling management of member data and family hierarchies.
- Designed a low-code backend using **n8n** on **Docker**, automating workflows for CRUD APIs, image compression, and ETL pipelines.
- Built an interactive analytics dashboard using **Recharts** to visualize demographic and employment data, supporting administrative planning.

Projects

Natural Language Data Explorer & Self-Healing Pipeline */ GitHub /*
Tech: Python, PySpark, Azure Databricks, Delta Lake, LangGraph, Streamlit Mar 2026 – Apr 2026

- Built a healthcare data-exploration prototype with **multi-agent Text-to-SQL** over synthetic **Synthea** data, including schema-drift handling and an AI schema advisor.
- Added role-based **PHI masking** and SQL safety checks, with a **Streamlit** interface and local tests for the schema-advisor logic.

Tool Stash */ GitHub /*
Tech: Python, FastAPI, React, TypeScript, Docker, Ollama Aug 2026 – Sep 2026

- Built an AI-assisted software-tool catalog that researches tools, drafts structured profiles for review, and supports browsing, import/export, and recommendations.
- Packaged with **Docker** Compose and CI-published multi-arch images; supports local **Ollama** and configurable hosted LLM providers.

Education

Indian Institute of Information Technology, Pune *2022-2026*
B.Tech, Computer Science and Engineering

Certifications & Achievements

- **Training:** AWS Cloud Practitioner Essentials; Boot.dev course certificates in Go, Docker, Linux, and SQL
- **Internship Certificate:** Artificial Intelligence with Python (Coincent); Postman API Fundamentals
- **Achievements:** Co-Head, R.O.F.I.E.S Robotics Club, IIIT Pune; 9th place, Vinyasa Summer of Code